MID TERM- MTH 641 — 5:30pm : 18th June' 2019

- For what norm $l^r$ space becomes Banach space.
- Separable space, dense set
- Theorem (continuous mapping) → {inverse image of open set is also open}.
- Theorem relevant to linear operator: (to show $T^{-1}$ is a linear operator)
  → isi se relevant ek MCQ bhi tha.
- MCQ's regarding completeness, compactness, finite dimensional space → open ball, closeness ?.
- How we can say that a subset is compact ?
- MCQ: Which one is not bounded operator

State Continuous Mapping Theorem 2

Define basis of finite dimensional vector Space. 2

Define Separable Space and dense set. 3

Write cannonical form of $R^n$ 3

Show that the space $l^\infty$ is complete 5

Given a norm space $(X, \|.\|)$,

Prove that $\big| \|X\| - \|Y\| \big| \leq \|X - Y\|$

Q Draw the unit sphere for the norm define by $\|x\|_2 = (|\xi_1|^2 + |\xi_2|^2)^{1/2}$ in $R^2$

Q Show that the space of all the bounded functions on a set $A$ is a metric space define as.

$$d(x,y) = \sup_{t \in A} |x(t) - y(t)|$$

Q show that the identity operator $I: X \to X$ on a normed space $X \neq \{0\}$ is bounded and has norm $\|I\| = 1$

Q Find countable subset $l^p$ for $1 \leq p < +\infty$

Q Define Riesz's Lemma.

Q Show rational numbers $Q$ are dens in $R$.

Note: Submit this sheet to Superintendent, before leaving the Examination Center.

Mth 641

22/6/2019 Current paper :-

1) Define the neighbourhood of a point in a metric space. (2)

2) Define a compact metric space X (2

3) Define convergent series in normed space (3)

4) Show that A subspace M is complete metric X is itself complete if the set M is closed in X. (3

5) Show that space $l^{\infty}$ is convergent (5

6) State prove compactness theorem. (5

[redacted]

MTH 641. Paper

Show that absolutely convergent series is normed space,
define complete metric space and....
define linear operator
Prove closure of M is......
X and Y are spaces then T:D(T)- - ->Y is linear operator if T^-1 exists if and only if Tx=0 implies that x=0
Prove Real line is a complete metric space

7:32 PM

Forwarded

MTH 641. Paper

1): Show that C is a Separable space,
2): compact space,
3): Give two examples of linear operator
4): Prove x belong to closure of M.
5): X and Y are spaces then T:D(T)- - ->Y is linear operator if T^-1 exists. Show that it is a linear operator. 6
6): Finite dimension prove

7:34 PM

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VU

**Rough Sheet**

FALL/SPRING MID/FINAL TERM 20__ (Year) Date: __/__/____ Superintendent Signature: [illegible]

Campus Code: VISD01 Student Signature: [illegible]

Course Code: MTH641 Student ID: [illegible]

(2) In a metric space $(X, d)$ every convergent sequence is Cauchy seq.

(3) The seq $\left(\frac{1}{n}\right)$ is strictly decreasing.

(9) A seq $(x_n)$ in a metric space is said to be strictly increasing if for all $n$, $x_n < x_{n+1}$

(10) Which of the following metric space is incomplete? Ans = space $\ell^p$

(6) Two metric spaces $(X, d)$ & $(\tilde{X}, \tilde{d})$ are said to be isometric if there exists an injective isometry of $X$ onto $\tilde{X}$

(12) Which of the following axioms holds for any vectors $u, v, w$ in vector space $X$, each in field $K$? Ans: $(u+v)+w = u+(v+w)$.

(8) Let $X$ be an $n$-dimensional vector space then any proper subspace $Y$ of $X$ has dimension equal to $n$.

(5) A sequence $(x_n)$ in a normed space $X$ is convergent if $X$ contains an $x$ s.t

(9) Let $T$ be a linear operator, and if $\dim D(T) = n < \infty$ then $\dim R(T) \leq n$

(3) $d(x, z) \leq d(x, y) + d(y, z) \Rightarrow$ Triangular property.

(4) $C[a, b]$ shows the space of all continuous functions defined on $[a, b]$

(6) $x \notin B(x_0; r) \Rightarrow d(x, x_0) \geq r$

(5) A discrete metric space $X$ is complete iff $X$ is countable.

(16) A metric space $X$ is said to be compact if every seq. in $X$ has a convergent subsequence

(17) Every finite dimensional normed space is Complete.

(18) Which of the following is not a bounded operator? $\Rightarrow$ Integral operator.

Note: Submit this sheet to Superintendent, before leaving the Examination Center

[redacted]

3. Show that $d(x, y) = \sqrt{|x-y|}$ defines a metric on the set of all real numbers.

**Solution:** Fix $x, y, z \in X = \mathbb{R}$, we need to verify the axioms of a metric. (M1) to (M3) follows easily from properties of absolute value. To verify (M4), for any $x, y, z \in \mathbb{R}$ we have

$$\begin{aligned}\Big[d(x,y)\Big]^2 = |x-y| &\leq |x-z| + |z-y| \\ &\leq |x-z| + |z-y| + 2\sqrt{|x-z|}\sqrt{|z-y|} \\ &= (\sqrt{|x-z|} + \sqrt{|z-y|})^2 \\ &= \Big[[d(x,z) + d(z,y)\Big]^2.\end{aligned}$$

Taking square root on both sides yields the triangle inequality.

[redacted]

[redacted]

5. Is boundedness of a sequence in a metric space sufficient for the sequence to be Cauchy? Convergent?

> **Solution:** It turns out that boundedness of a sequence in a metric space does not imply Cauchy or convergent. Consider the sequence $(x_n)$, where $x_n = (-1)^n$; it is clear that $(x_n)$ is bounded in $(\mathbb{R}, d)$, where $d(x_m, x_n) = |x_m - x_n|$.
>
> - $(x_n)$ is not Cauchy in $\mathbb{R}$ since if we pick $\varepsilon = \dfrac{1}{2} > 0$, then for all $N$, there exists $m, n > N$ (we could choose $m = N+1$, $n = N+2$ for example) such that $d(x_m, x_n) = d(x_{N+1}, x_{N+2}) = 1 > \dfrac{1}{2}$.
> - $(x_n)$ is not convergent in $\mathbb{R}$ since it is not Cauchy.

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VU

SPRING MID TERM 2019

Campus Code: [illegible]

Course Code: MTH641

Rough Sheet

Date: [illegible]

Superintendent Signature: [illegible]

Student Signature: [illegible]

Student ID: [illegible]

① Define Isometric mapping write the usual metric "d" in the Euclidean plan $R^2$ and Euclidean space $R^3$

② write the norm on $R^n$ for which it is a Banach space

③ If $T$ is a linear operator then show that the null space $N(T)$ is a vector space ⑤

④ Prove that $x \in \bar{M}$ if there is a sequence $(x_n)$ in $M$ such that $x_n \to x$ ⑤

⑤ Let $X$ be an n-dimensional vector space. Then show that any proper subspace $Y$ of $X$ has dimension less than $n$.

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Q1- $B(x_0, 1) < r$. Justify its elements

Q2. Define series in normed spaces.

Q3. Differentiate between finite dimensional vector space and infinite dimensional vector space.

Q4. Prove that every convergent series in a metric space is a Cauchy sequence.

Q5. Prove $d(x,y) = \sqrt{x-y}$ triangle inequality

Q6. Prove that $\|\cdot\|$ is equivalent to $\|\cdot\|_0$.

"Current Paper"

(MTH-641) 21-June-2019

1. Definition of Accumulation point. (2)
2. Prove that Real line is metric space (2)
3. Give 4-examples of linear operator. (3)
4. Give 5-examples of norm space. (3)
5. Prove that $(ST)^{-1} = T^{-1}S^{-1}$ (5)
6. Theorem prove krna tha continuous mapping ka. (5)

* MCQ's ~~[illegible]~~ (very easy)

Mostly from theorems and Lemma's statements.

Norm space ki 5 examples

1:04 PM

D (3,2) ki vector spaces find krni thi

1:05 PM

ST =TS inverse prove krna tha

1:06 PM

Deff of continuous 1:07 PM

Continuous wala theorm 1:07 PM

Define  convergence in normed space.          Differetiate between finite dimisional vector space anf

12:13 pm

And infiite dimensional space,prove that every sequence convergence in a metric space is a cauchy sequence.

12:13 pm

Mth641 pppr 12:14 pm

Mth641 my paper

Define series of normed space
Give 5 examples of normed space
Is boundends of a sequence in a metric space sufficient for the sequence to be Cauchy? or convergent?
Show that d(x,y)=|x-y|underoot defines a metric  on the set of all real numbers

mth641 today's paper
triangle inequality prove krni thi.(5)
bounded linear operator define.(3)
cauchy sequency ki convergent btani thi xn=1+1/n ki(2)
linear operator ki 4 examples (2)
compactness ka theorem tha.(5)mujy nai ata tha sai wala
isometric ki defination (3)